

Improving Generalization of Deepfake Detectors by Imposing Gradient Regularization

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Outline

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What are Deepfakes?



- Deepfakes are synthetic media generated by AI models (e.g., GANs), capable of producing realistic faces and expressions.
- These videos are often indistinguishable from real ones to the human eye.

Why is Detection important?



Disinformation

Fake videos spread misinformation, manipulate public opinion, and undermine democracy.



Defamation

Famous people can be falsely shown in compromising contexts, damaging their image.



Trust

Reliable detectors protect media integrity and public confidence in digital content.

Related Works

CNN-based Deepfake Detection

- Use convolutional architectures trained on visual features (e.g., EfficientNet) [2, 3]
- Performance drops under perturbations

Attack Techniques:

- Methods to subtly alter images and fool the model [5, 6, 7]

Robustness Techniques

- Adversarial Training: [6]
- Gradient Regularization: enhances generalization and smoothness [1]
- Evaluation protocols highlight robustness under attack [4]

Proposed Methods 1

In order to improve the robustness and generalization of our detector, we employ two complementary regularization techniques:

- **Perturbation Injection Module (PIM)**
 - Simulates distribution shifts by perturbing shallow feature statistics through a regularization term added to the loss function.

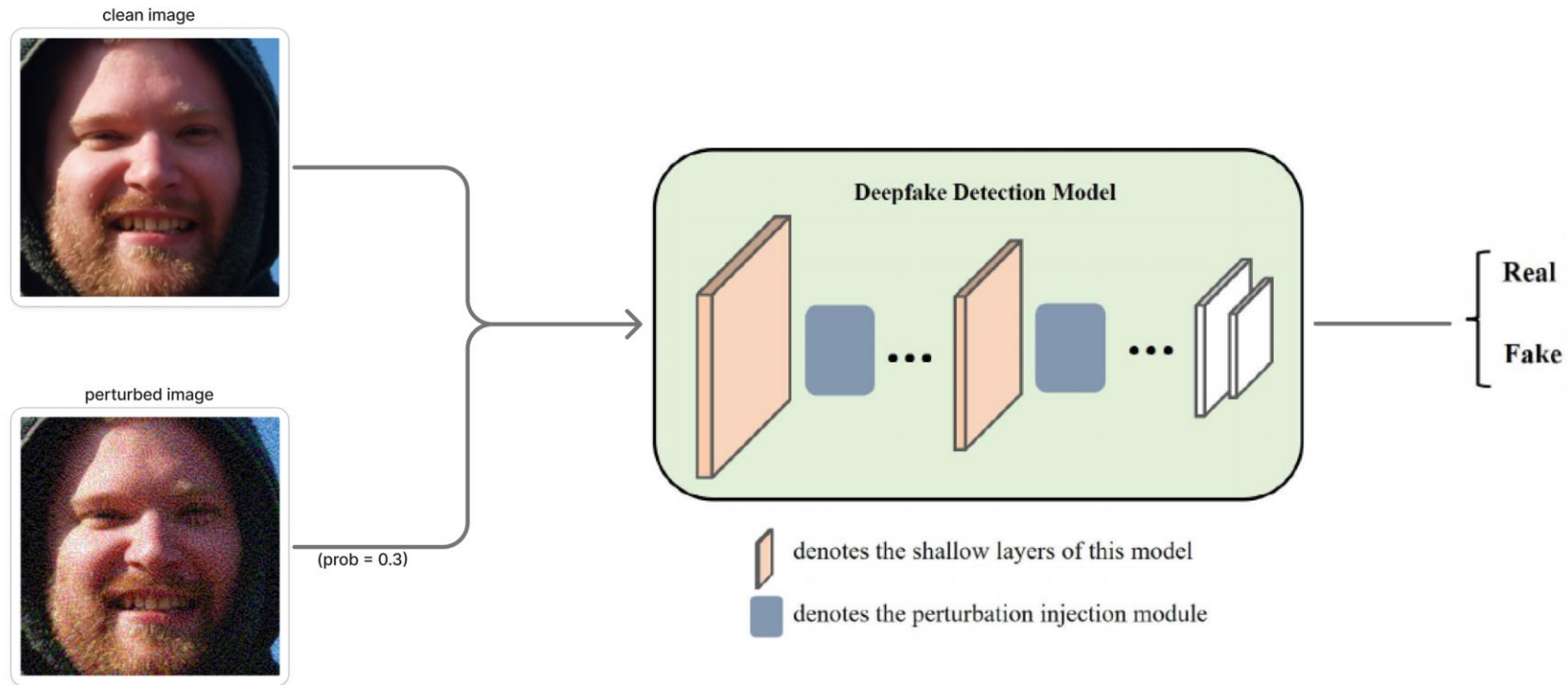
$$\min_{\theta} (1 - \alpha) L(x, y, \theta) + \alpha L(f'_{\theta_s}(x), y, \theta_d)$$

- **Adversarial Training**
 - Improves resistance to input perturbations using adversarial examples in the training step

$$x_{t+1} = \Pi_{\mathcal{X}+\mathcal{S}} (x_t + \alpha \cdot \text{sign}(\nabla_x \mathcal{L}(\theta, x, y)))$$

These strategies are tested independently and in combination.

Proposed Methods 2

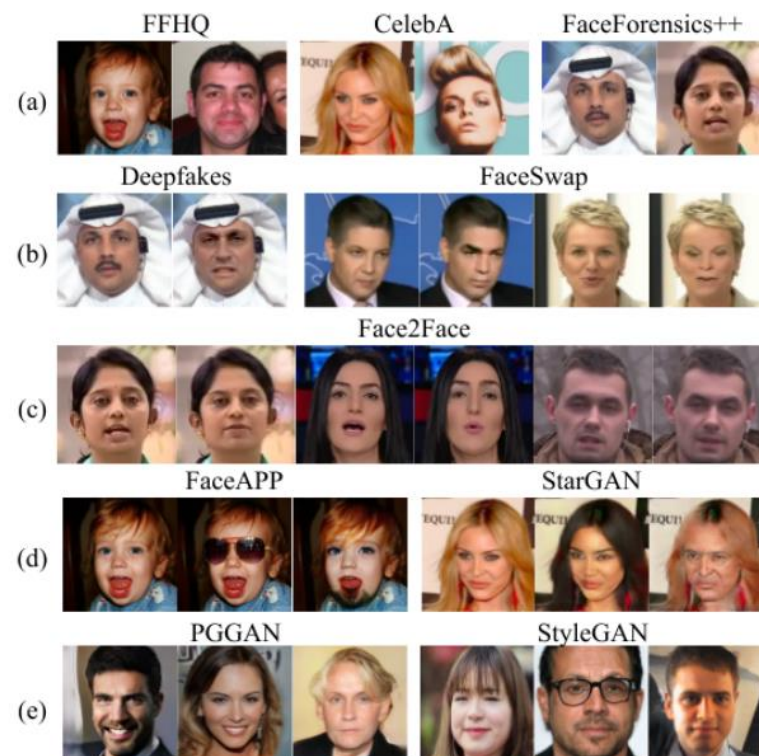


Adversarial Attacks

To evaluate model robustness, we tested all configurations against four white-box adversarial attacks:

- **FGSM (Fast Gradient Sign Method)**
Adds a single-step perturbation in the direction of the gradient sign.
- **I-FGSM (Iterative FGSM)**
Applies FGSM multiple times with small steps and clipping at each iteration.
- **DeepFool**
Optimization-based attack that computes the minimal perturbation required to change the model's prediction.
- **PGD (Projected Gradient Descent)**
Strongest first-order adversary; iteratively adds perturbations and projects into an ϵ -ball.

DFFD: Diverse Fake Face Dataset



Evaluation Metrics

- **Accuracy:** Measures the proportion of correctly classified examples over the total

$$\frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** Indicates the fraction of predicted positives that are actually positive

$$\frac{TP}{TP + FP}$$

- **Recall:** Measures the model's ability to identify all actual positives

$$\frac{TP}{TP + FN}$$

- **F1:** Harmonic mean of precision and recall

$$2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

- **Attack Robustness:** Measures the model's resistance to adversarial attacks

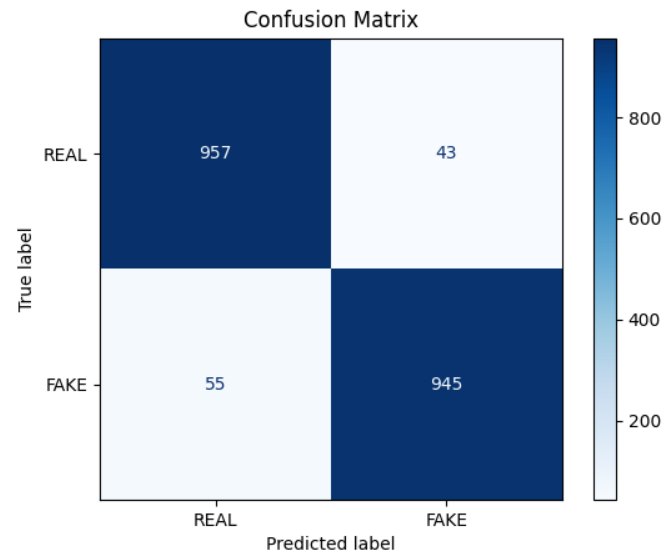
$$1 - \frac{\#flipped\ predictions}{\#total\ samples}$$

Attack success rate

Baseline Evaluation

We first fine-tune the EfficientNet-B0 baseline model on our dataset to create a reference point.

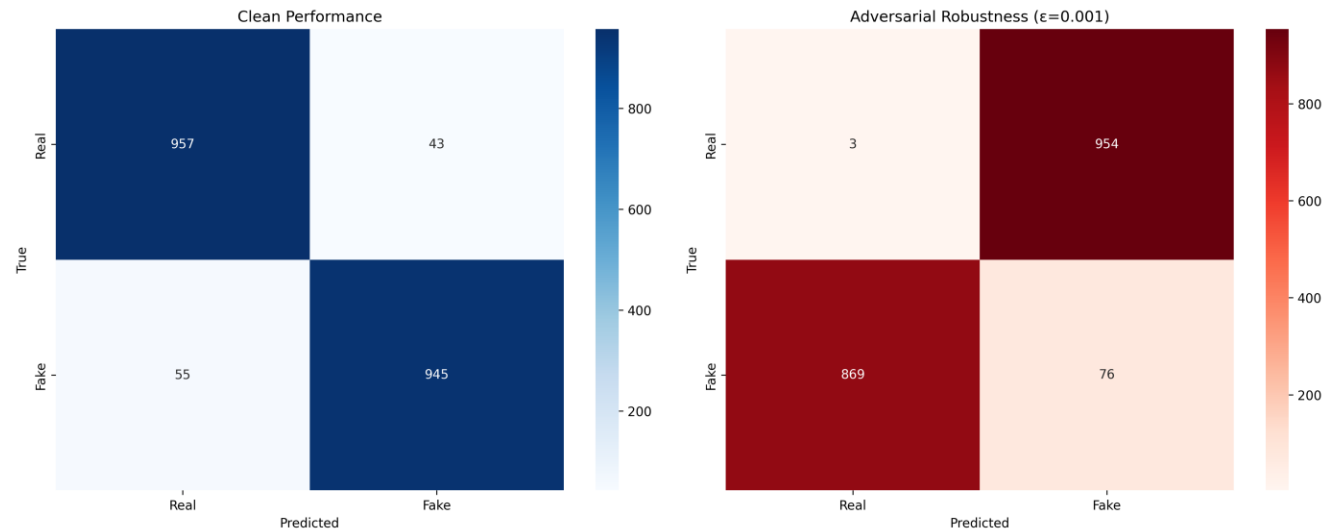
- The results demonstrate that the model achieves **high performances on clean data**.



Class	Precision	Recall	F1-Score
0 (Real)	0.9457	0.9570	0.9513
1 (Fake)	0.9565	0.9450	0.9507
Avg	0.9511	0.9510	0.9510

Adversarially Attacked Model Evaluation

- Despite achieving high accuracy on clean data, the model's performance **drops significantly** when subjected to even small perturbations, revealing its **lack of robustness** to adversarial inputs.
- This highlights the **need for targeted robustness strategies** to ensure reliability in real-world scenarios.



Robustness Evaluation

- Under adversarial training, PIM improves robustness consistently across most attacks.
- These results show that our methods make the model much more resistant to adversarial attacks, helping it perform more reliably in real-world situations.

Attack	ϵ	PIM	Non-PIM	Gain
Adversarial Training				
FGSM	0.001	49.69	32.27	+17.42
FGSM	0.01	40.34	13.00	+27.34
FGSM	0.1	51.71	49.09	+2.62
FGSM	0.5	46.16	49.61	-3.45
I-FGSM	0.001	28.87	8.25	+20.62
I-FGSM	0.01	5.19	4.91	+0.28
I-FGSM	0.1	82.19	71.07	+11.12
I-FGSM	0.5	95.43	84.65	+10.78
PGD	0.001	30.06	8.46	+21.60
PGD	0.01	5.50	5.38	+0.12
PGD	0.1	84.53	75.14	+9.39
PGD	0.5	98.29	91.33	+6.96
DeepFool	–	45.90	34.93	+10.97

Table 1: Robustness (%) under adversarial training.

Attack	ϵ	PIM	Non-PIM	Gain
Normal Training				
FGSM	0.001	15.69	4.15	+11.54
FGSM	0.01	11.06	3.42	+7.64
FGSM	0.1	50.80	53.58	-2.78
FGSM	0.5	50.90	46.58	+4.32
I-FGSM	0.001	8.99	0.00	+8.99
I-FGSM	0.01	8.72	0.00	+8.72
I-FGSM	0.1	83.03	53.31	+29.72
I-FGSM	0.5	88.94	75.76	+13.18
PGD	0.001	9.04	0.00	+9.04
PGD	0.01	0.00	0.00	+0.00
PGD	0.1	86.54	57.99	+28.55
PGD	0.5	89.47	88.01	+1.46
DeepFool	–	9.31	19.40	-10.09

Table 2: Robustness (%) under normal training.

Conclusion and Future Work

- Our findings highlight that the Perturbation Injection Module (PIM), when combined with adversarial training, significantly improves the robustness of deepfake detection against a wide range of adversarial attacks.
- Future research could focus on extending our evaluation to include a wider range of deepfake generation techniques and more challenging datasets. It would also be valuable to investigate whether combining PIM and adversarial training with other defense strategies can further enhance robustness and generalization.

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